

Distilled but Not Private: Investigating Model Inversion Attacks on IoT Intrusion Detection Systems

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Abstract. Knowledge distillation (KD) is a common technique for deploying intrusion detection models on resource-constrained Internet of Things (IoT) devices, yet its effect on training-data leakage remains unexplored. In this paper, we investigate the risk of reconstructing the teacher’s private training data via the student with model inversion attacks (MIAs) under a deployment-relevant setting where the adversary has white-box access to the student and the proprietary teacher remains protected. Across four public tabular IDS datasets and varying the distillation objective, we find that: (i) posterior-based inversion consistently outperforms the logit-based variants in this tabular IDS context (+14.79% accuracy on average); (ii) ground-truth-supervised distillation is the dominant factor of leakage, increasing inversion success by 20.58% relative to teacher-only KD; and (iii) within the evaluated MLP family, teacher-student architectural mismatch has limited impact compared to the distillation objective. These findings highlight privacy risks when releasing distilled IoT NIDS models and suggest that privacy-aware objectives should be a focus for KD-based NIDS design.

Keywords: knowledge distillation · model inversion attack · privacy leakage · internet-of-thing · intrusion detection

1 Introduction

The rapid growth of Internet of Things (IoT) devices has challenged traditional signature-based network intrusion detection systems (NIDS), driving the emergence of machine-learning-powered NIDS. Traditional NIDSs, which primarily rely on signature and rule databases, have become increasingly difficult to maintain as these databases grow in proportion to the number of signatures [10,4]. Moreover, such systems fail to detect zero-day attacks whose signatures are absent from the databases unless the attacks are carefully analyzed to construct corresponding signatures [23,9]. This process often requires expertise and substantial time, hindering traditional NIDSs from detecting novel threats in real time.

As a result, machine-learning-powered systems have been increasingly adopted because they can automatically learn complex network patterns while adapting quickly to evolving threats [3,15,21,1].

However, deploying large detection models on IoT systems is challenging because IoT devices have limited computational resources. This constraint motivates the development of model compression techniques to reduce the footprint of large models while maintaining detection accuracy. To this end, knowledge distillation (KD) [8] is a promising approach that transfers knowledge from a large teacher model to a smaller student model, thereby reducing computational requirements while preserving detection accuracy. Consequently, KD has been widely applied in IoT NIDS to detect a wide range of threats [11,17].

Despite its promising applications, KD poses privacy risks to the teacher’s confidential training data. By transferring knowledge from a teacher to a student model, KD may expose sensitive information from the teacher’s training set to adversaries through the student model. Recent research [20,16] showed that KD can transfer richer information from the training set to the student model than training the student on such data alone. The reason is that KD transfers both information about the training data and the correlations that the teacher has discovered (i.e., dark knowledge). Furthermore, recent work on model inversion attacks (MIAs) [13,6,27] has leveraged KD to reconstruct confidential training data from a black-box teacher model. This leads to detrimental consequences for IoT NIDS: adversaries may reconstruct sensitive training data via the student model, which often includes confidential data from industrial systems or health records, thereby risking violations of data privacy regulations (e.g., GDPR [18]).

Although the feasibility of reconstructing a teacher’s confidential training data via a student model has been widely demonstrated, existing work primarily focuses on vision domains. As a result, it remains unclear whether similar privacy risks arise in tabular domains—especially in IoT NIDS—because feature semantics, distributions, and constraints differ substantially from images. In particular, tabular network data exhibits strong logical and semantic correlations across features and across samples. For example, a distributed denial-of-service (DDoS) attack may generate a high volume of traffic (i.e., many records) that share the same protocol, destination IP address, and port. If one considers only a single feature (e.g., protocol) or a single record, it is unlikely to reveal the DDoS attempt. These characteristics motivate our investigation into whether MIAs remain effective in tabular settings.

In this paper, we investigate the privacy risks associated with deploying distilled student models in IoT NIDS. Specifically, we examine whether MIAs can reconstruct a teacher’s private training data through the student model. We focus on Multi-Layer Perceptron (MLP) models and four popular tabular IDS datasets (IoTID20, CIC-IDS-2017, CIC-IDS-2018, and CIC-IoT-2023). We consider two KD methods—*Traditional KD* [8] and *Teacher-only KD* [20]—and two MIA methods—*Traditional MIA* [7] and *Improved MIA* [14]. We assume the following threat model: the adversary has white-box access to the student model

but no access to (or knowledge of) the teacher model or its private training data. Under this setting, we investigate the following research questions (RQs):

- **RQ1:** How effective are MIA methods against different MLP models?
- **RQ2:** How effective are MIA methods against different KD methods?
- **RQ3:** What is the impact of teacher–student architectures on MIA effectiveness?

In RQ1, we first study the intrinsic vulnerability of the teacher model itself to MIA, and whether this vulnerability depends on the teacher MLP architecture. This baseline is essential before studying privacy leakage from KD: if different teachers leak at different rates, then any privacy claims about KD must be interpreted relative to the teacher’s inherent leakage. Building on this baseline, RQ2 delves into the primary drivers of privacy leakage by contrasting Traditional KD with Teacher-Only KD. Our goal is to untangle the vulnerability introduced by ground-truth supervision from the inherent transfer of dark knowledge. Uncovering this distinction provides actionable insights for privacy-preserving IoT NIDS, revealing whether optimizing the distillation objective alone is sufficient to thwart model inversion attacks without compromising model utility. Finally, RQ3 examines whether teacher–student architectural discrepancies further amplify or mitigate inversion risk. We evaluate MIA effectiveness using attack accuracy and K-nearest neighbors (KNN) distance [7], along with an additional attack utility metric. Attack accuracy and KNN distance quantify the fidelity between reconstructed samples and the original private training data, whereas attack utility measures the usefulness of the reconstructed data for downstream ML tasks.

Overall, our results answer the three research questions as follows. For RQ1, across both deep and baseline MLP teachers, the Traditional MIA is consistently more effective than the Improved (logit-based) MIA method in the evaluated tabular IDS setting, indicating that logit-based inversion does not provide a reliable advantage. In RQ2, the distillation method is indeed a driver of privacy leakage. In detail, the Traditional KD produces substantially more vulnerable student models than the Teacher-Only KD due to the additional ground-truth supervision from the Traditional KD. Finally for RQ3, within the evaluated MLP family, the teacher–student architectural mismatch has a limited influence on the attack effectiveness. Specifically, when distilling four student variants, attack accuracy varies within a narrow margin. These results suggest that the choice of KD objective dominates architecture-level changes in determining the risk of data leakage in KD.

The remainder of this paper is organized as follows. Section 2 presents related work. Section 3 describes the evaluated KD and MIA methods. Sections 4 and 5 detail the experimental setup and results. Finally, Sections 6 and 7 discuss the limitations of our study and conclude the paper.

2 Related Work

KD is a model compression technique that transfers knowledge from a large teacher model to a smaller student model, originally introduced by Hinton et al. [8]. In resource-constrained environments such as IoT NIDS, KD has been widely used to deploy lightweight models while maintaining strong detection performance [26,25]. Beside reducing the model’s footprint, KD exhibits a risk of privacy leakage because the student is trained to match the teacher’s softened predictions, which encode so-called *dark knowledge*. According to the systematic studies in [20,16], students can inherit teacher behaviors that are not explicitly targeted by the distillation objective. For example, students can transfer the teacher’s *localization* behavior (their class-activation maps become more similar to the teacher’s than those of independently trained students), and teacher’s *invariances to data augmentations* (a color-jitter-robust teacher can yield a student that is also robust against color jitters even if the student is distilled without color augmentations). As a result, the student can serve as an information-rich proxy about the teacher’s learned knowledge and its private training set, posing critical privacy risks when the student is deployed in sensitive environments like IoT.

MIAs provide a mechanism for realizing such privacy leakage. In detail, MIAs exploit the model’s input–output mapping to infer sensitive properties or reconstruct representative training samples [5]. In a seminal work, Fredrikson et al. [7] demonstrated that confidence scores can be abused to recover sensitive attributes. Subsequent work has proposed diverse inversion strategies, including approaches that leverage auxiliary data and generative modeling to improve reconstruction fidelity [2,28,24]. However, Nguyen et al. [14] argued that standard likelihood inversion objectives yield reconstructed samples that minimize the loss while visible representations significantly differ from the private training data. Consequently, they proposed modifying the inversion loss to better align optimization with semantic reconstruction quality, leading to improved inversion accuracy, lower KNN distances, and higher attack utility. Recent studies further suggest that MIA risks can be exacerbated when KD is used to construct a substitute. For instance, label-only inversion settings can be strengthened by training a surrogate model via KD and then performing a white-box attack against the surrogate model [13]. While these advances highlight the growing capability of MIAs to reconstruct private training data, existing studies have not examined how such attacks interact with knowledge distillation in tabular domains, especially IoT NIDS. As a result, it remains unclear whether distilled student models in IoT NIDS leak sensitive teacher knowledge under MI attacks, which motivates our study.

3 Methodology

This section presents the KD and MIA methods evaluated in our study.

3.1 Knowledge Distillation Methods

We consider two KD methods. The first follows the original work of Hinton et al. [8], which we refer to as **Traditional KD**. The second is inspired by the systematic study of Stanton et al. [20], which we refer to as **Teacher-only KD**.

Traditional KD. In this KD method, the student model is trained to minimize a joint loss function $\mathcal{L}_{\text{Traditional}}$, defined as a weighted combination of a task loss and a distillation loss:

$$\mathcal{L}_{\text{Traditional}} = \alpha \mathcal{L}_{\text{task}} + (1 - \alpha) T^2 \mathcal{L}_{\text{distillation}}, \quad (1)$$

where $\alpha \in [0, 1]$ weights the two loss terms, and T is a temperature hyperparameter used to soften the teacher and student probability predictions.

In neural network, class probabilities are typically derived using a softmax layer that maps the logit z_i of each class to a probability $p(z_i, T)$ by normalizing it over the logits of all classes. We define:

$$p(z_i, T) = \frac{\exp(z_i/T)}{\sum_{c=1}^C \exp(z_c/T)}, \quad (2)$$

where C is the number of classes.

Let $z^s, z^t \in \mathbb{R}^C$ denote the pre-softmax outputs (logits) of the student and teacher models for an input x . Then, $\mathcal{L}_{\text{task}}$ is the cross-entropy loss between the student's softmax output (with $T = 1$) and the ground-truth label:

$$\mathcal{L}_{\text{task}} = - \sum_{i=1}^C y_i \log p(z_i^s, 1), \quad (3)$$

where y is a one-hot vector representing the ground-truth label of x .

Subsequently, $\mathcal{L}_{\text{distillation}}$ is the Kullback–Leibler (KL) divergence between the teacher and student softmax outputs:

$$\mathcal{L}_{\text{distillation}} = \mathcal{D}_{KL} (p(z^t, T) \parallel p(z^s, T)) = \sum_{i=1}^C p(z_i^t, T) \log \left(\frac{p(z_i^t, T)}{p(z_i^s, T)} \right). \quad (4)$$

Teacher-Only Knowledge Distillation. Teacher-Only KD does not include the supervised task loss $\mathcal{L}_{\text{task}}$; hence, the student is not directly trained to fit ground-truth labels. To this end, the loss function of this Teacher-Only KD is defined as:

$$\mathcal{L}_{\text{TeacherOnly}} = T^2 \mathcal{L}_{\text{distillation}}. \quad (5)$$

This method is useful for evaluating privacy risks in KD because Teacher-Only KD transfers *dark knowledge* from the teacher to the student without using ground-truth supervision, unlike Traditional KD. In our experiments, Teacher-Only KD uses the same temperature T as Traditional KD and reuses the same KL-based distillation loss definition..

3.2 Model Inversion Attack Methods

We study two seminal MIA methods. The first was proposed by Fredrikson et al. [7], which we refer to as **Traditional MIA**. The second, introduced by Nguyen et al. [14], is referred to as **Improved MIA** because it improves upon Traditional MIA. Notably, we subsequently use the term *target model* to denote the model under attack: it refers to the *teacher* when MIA is conducted against the teacher, and to the distilled *student* when MIA is conducted against the student. Both methods share a common optimization process (see Algorithm 1) and differ only in the objective function $\mathcal{L}_{id}$.

Algorithm 1 Optimization-based Model Inversion Attack

Require: Target model $f(\cdot)$ with fixed parameters, target label y

Ensure: Reconstructed input $\hat{x}$

- 1: Initialize $x_0 \sim \mathcal{N}(0, 0.1^2)$
 - 2: **for** $i = 1$ to $I_{\max}$ **do**
 - 3: Compute inversion loss $\mathcal{L}_{id}(x_{i-1}; y)$
 - 4: Update input: $x_i \leftarrow x_{i-1} - \eta \nabla_x \mathcal{L}_{id}(x_{i-1}; y)$
 - 5: Apply feature constraints and normalization
 - 6: **end for**
 - 7: **return** $\hat{x} = x_{I_{\max}}$
-

Traditional MIA. This method reconstructs an input by maximizing the posterior probability of a target class under the target model. Let $z^{\text{tar}}(x) \in \mathbb{R}^C$ denote the pre-softmax output (logits) of the target model for input x , where C is the number of classes. The posterior probability of class y is defined as:

$$P(y \mid x) = \frac{\exp(z_y^{\text{tar}}(x))}{\sum_{c=1}^C \exp(z_c^{\text{tar}}(x))}. \quad (6)$$

Formally, the attack solves the following optimization problem:

$$\hat{x} = \arg \max_{x \in \mathcal{X}} P(y \mid x), \quad (7)$$

which is equivalently expressed as minimizing the negative log-likelihood:

$$\mathcal{L}_{id}(x; y) = -\log P(y \mid x). \quad (8)$$

Improved MIA. Instead of minimizing the loss function in Equation 8—for which the optimizer may reduce the objective by increasing the softmax denominator rather than increasing the numerator—Improved MIA replaces posterior maximization with a logit-maximization strategy.

Let $h(x)$ denote the output of the penultimate layer of the target model for input x , and let w_y denote the weight vector of the last linear layer corresponding to class y of the target model. The logit of class y is defined as:

$$z_y^{\text{tar}}(x) = w_y^\top h(x), \quad (9)$$

and the inversion objective is:

$$\mathcal{L}_{id}^{\text{logit}}(x; y) = -z_y^{\text{tar}}(x) + \lambda \|h(x) - h_{\text{reg}}\|_2^2, \quad (10)$$

where λ is a regularization coefficient, and h_{reg} denotes a class-conditional feature prior for class y . Since the attacker has no access to the target model’s private training data, we estimate h_{reg} using a small auxiliary dataset from the same domain. Specifically, we pass auxiliary samples from class y through the target model, extract their penultimate-layer features, and use the mean feature vector as h_{reg} . This practice aligns with the original paper of the Improved MIA method [14].

Implementation Details and Hyperparameters For both attack variants, reconstruction is performed via iterative gradient-based optimization in the input space with fixed target model parameters. The input is initialized as Gaussian noise $x_0 \sim \mathcal{N}(0, 0.1^2)$. We optimize x using SGD with an initial learning rate of 0.005 and a momentum of 0.9 for at most 3,000 iterations ($I_{\text{max}} = 3000$). Early stopping is applied if the inversion loss does not decrease for 100 consecutive iterations or drops below 0.05. At each iteration, the reconstructed input is normalized and clipped to $[-1, 1]$. For Improved MIA, the regularization coefficient is set to $\lambda = 0.1$.

4 Experimental Setup

4.1 Dataset

Our study considers four IDS datasets: IoTID20, CIC-IDS-2017, CIC-IDS-2018, and CIC-IoT-2023. IoTID20 [22] consists of network traffic from two smart-home devices: SKT NUGU (NU 100) and an EZVIZ Wi-Fi camera (C2C Mini O Plus 1080P). We preprocess the raw traffic from the original PCAP files and select a subset of 261,538 samples from five classes, removing highly under-represented classes. CIC-IDS-2017 [19] contains labeled network flows extracted with CICFlowMeter, covering benign traffic and several attack types. For this dataset, we use 133,699 samples across eight classes after filtering incomplete flows and removing under-represented attack classes. CIC-IDS-2018 [19] is a large-scale, real-world network traffic dataset covering a wide range of attacks, including DDoS, botnet, and brute-force attacks. We select a subset of 253,050 samples from eight classes, with an equal number of samples per class. Finally, CIC-IoT-2023 [12] was collected from an extensive topology involving real IoT devices acting as attackers and victims across 33 attack scenarios. We use 341,210 samples across four attack categories, which merge finer-grained attack classes with similar behaviors. Table 1 summarizes the evaluated datasets.

Dataset	Samples	Features	Number of Classes
CIC-IDS-2017	133,699	81	8
CIC-IDS-2018	253,050	79	8
CIC-IoT2023	341,210	47	4
IoTID20	261,538	80	5

Table 1. Summary of datasets.

4.2 Model Architectures

We focus on the Multi-Layer Perceptron (MLP) architecture, as it is one of the most popular and simplest neural network architectures for tabular data. The network consists of a sequence of blocks, each comprising a linear layer, a batch-normalization layer, and a ReLU activation function. The architecture begins with an initial block that maps the input features to a specified hidden dimension, followed by a number of hidden blocks that transform the data while preserving this dimension. Finally, a linear layer maps the output of the last hidden block to the classification logits. Based on this structure, we define a **baseline** MLP with two hidden blocks, each with a width of 1024. Other variants are obtained by varying the hidden dimension and the number of hidden blocks, as described in the corresponding RQs.

4.3 Preprocessing

We use the Scikit-learn³ and Imbalanced-learn⁴ libraries to implement the following data preprocessing pipeline:

Data Loading and Splitting: The raw dataset is first loaded into a Pandas `DataFrame` and then split into a training set (80%) and a test set (20%).

Feature Transformation: Features in the training set undergo a three-step transformation process:

- **Imputation:** Missing values are imputed with the mean value of the corresponding columns.
- **Quantile Transformation:** Features are transformed to follow a normal distribution using `QuantileTransformer`⁵.
- **Scaling:** Transformed features are scaled to the range $[-1, 1]$ using `MinMaxScaler`⁶.

Label Processing: Missing labels are imputed using the most frequent value in the training data. Subsequently, categorical labels are converted into numerical values using a `LabelEncoder`⁷.

³ <https://scikit-learn.org/>

⁴ <https://imbalanced-learn.org/>

⁵ <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.QuantileTransformer.html>

⁶ <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html>

⁷ <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.LabelEncoder.html>

Handling Class Imbalance: To address class imbalance in the training data, we apply `RandomUnderSampler`⁸, which randomly downsamples majority classes to match the sample size of the minority class. This aims to mitigate the severe class imbalance commonly observed in IDS datasets, which can cause the model to overfit to majority classes. Notably, we focus on the balanced-data setting, leaving the evaluation of MIA under naturally imbalanced data distributions for future work.

To avoid information leakage from the training set to the test set, the above preprocessing procedure is fitted exclusively on the training set, and the resulting transformers, scalers, encoders, and samplers are then applied to the test set.

4.4 Evaluation Metrics

Following common practices in MIA [13], we evaluate attack effectiveness using three metrics:

Attack Accuracy: We quantify reconstruction fidelity using an independent evaluation model. This model is distinct from the target (teacher or student) model and is trained on the same private training set. Attack accuracy is defined as the classification accuracy of the evaluation model on reconstructed samples with their intended labels. Higher attack accuracy indicates that reconstructed samples preserve discriminative characteristics of the private training data.

K-Nearest Neighbors Distance (KNN Dist): This metric measures the mean Euclidean distance between each reconstructed sample and its K nearest samples from the same class in the teacher’s private training set (in input space). We report the average of these distances over all reconstructed samples.

Attack Utility: Finally, while attack accuracy and KNN distance provide analytic measures of fidelity, we also quantify the practical utility of reconstructed data for downstream ML tasks. Specifically, we train a new model solely on the reconstructed data and evaluate its accuracy on the teacher’s held-out test set. This evaluates whether the attack recovers feature-label dependencies that generalize to real traffic, i.e., whether reconstructed records can serve as a substitute for private training data in downstream IDS development. To provide a comprehensive evaluation, we consider four downstream models:

- **A shallow MLP (mlp1hidden):** This MLP consists of a single hidden block with a dimension of 128.
- **A deep MLP (mlp4hidden):** This MLP has four hidden blocks with decreasing dimensions (3000, 2000, 1000, and 500, respectively).
- **A Random Forest classifier:** We use `RandomForestClassifier` from the `Scikit-learn` library with 100 decision trees.
- **An XGBoost classifier:** We use `XGBClassifier` from the `XGBoost` library with 100 boosting rounds and a maximum depth of 4.

⁸ https://imbalanced-learn.org/stable/references/generated/imblearn.under_sampling.RandomUnderSampler.html

5 Experimental Results

We structure our experiments with three research questions (RQs):

- **RQ1:** How effective are MIA methods against different MLP models?
- **RQ2:** How effective are MIA methods against different KD methods?
- **RQ3:** What is the impact of teacher-student architectures in KD on the effectiveness of MIA?

5.1 RQ1: Impact of different MLP models on MIA effectiveness

Table 2. Comparison of MIA methods against different MLP teacher models.

Model / Attack	Attack Accuracy (%)	KNN	Attack Utility (%)	
		Distance	mlp1hidden	mlp4hidden
Deep – Traditional MIA	59.35 $\pm$ 28.56	5.93 $\pm$ 0.43	35.48 $\pm$ 2.23	36.60 $\pm$ 4.18
Deep – Improved MIA	44.57 $\pm$ 12.08	5.98 $\pm$ 0.46	32.73 $\pm$ 5.17	32.33 $\pm$ 8.98
Baseline – Traditional MIA	74.60 $\pm$ 28.80	5.84 $\pm$ 0.43	37.83 $\pm$ 2.70	36.79 $\pm$ 5.53
Baseline – Improved MIA	33.20 $\pm$ 40.42	5.77 $\pm$ 0.29	29.37 $\pm$ 0.04	28.02 $\pm$ 0.06
Narrow – Traditional MIA	58.44 $\pm$ 38.34	5.73 $\pm$ 0.37	38.11 $\pm$ 4.25	39.91 $\pm$ 6.71
Narrow – Improved MIA	66.11 $\pm$ 33.05	5.82 $\pm$ 0.46	29.81 $\pm$ 0.05	27.56 $\pm$ 0.07

RQ1 establishes a *teacher-only* baseline by measuring the effectiveness of MIAs when applied *directly* to the teacher model (i.e., without distillation). This baseline contextualizes later experiments in which MIAs target distilled student models by isolating the teacher’s intrinsic privacy leakage from any additional leakage introduced by KD.

Concretely, we evaluate two inversion methods (Traditional and Improved) against three teacher MLP architectures trained on the *same* training set: (i) a **baseline** MLP, (ii) a **deep** MLP with four hidden blocks of width 1024, and (iii) a **narrow** MLP with two hidden blocks of width 512. We report attack accuracy, KNN distance, and attack utility (Section 4.4).

Table 2 shows that direct MIA effectiveness depends on teacher architecture. For Traditional MIA, the **narrow** MLP is the most vulnerable, achieving the lowest KNN distance (5.73 ± 0.37) and the highest downstream attack utility (38.11 ± 4.25 for **mlp1hidden** and 39.91 ± 6.71 for **mlp4hidden**), whereas the **deep** MLP is harder to invert. Improved MIA underperforms Traditional MIA, particularly in downstream utility, across all teachers. Overall, these results indicate that teacher architecture affects privacy leakage and that Traditional MIA provides a reliable teacher-only leakage baseline for subsequent KD comparisons.

5.2 RQ2: Impact of different KD methods on MIA effectiveness

We next study the impact of the two KD methods on MIA effectiveness. RQ2 investigates whether the distillation objective itself influences privacy leakage under model inversion attacks. In Traditional KD, the student is optimized using

Table 3. Comparison between both KD methods regarding MIA effectiveness.

Metric	Traditional KD (mean $\pm$ std)	Teacher-Only KD (mean $\pm$ std)	Comparison (Trad. KD - T-Only KD)
Attack Accuracy (%)	59.19 $\pm$ 5.53	38.61 $\pm$ 3.87	+20.58
KNN Distance	5.8269 $\pm$ 0.0012	5.8269 $\pm$ 0.0004	0.00
Attack Utility (%) - mlp1hidden	26.44 $\pm$ 0.44	26.10 $\pm$ 0.32	+0.34
Attack Utility (%) - mlp4hidden	25.67 $\pm$ 0.89	26.01 $\pm$ 0.70	-0.34
Attack Utility (%) - random forest	22.21 $\pm$ 0.11	22.34 $\pm$ 0.09	-0.13
Attack Utility (%) - xgboost	24.31 $\pm$ 0.11	24.33 $\pm$ 0.10	-0.02

both the teacher’s softened predictions and the ground-truth labels, whereas Teacher-Only KD relies solely on the teacher’s softened outputs. Consequently, this experiment is set up not merely to compare KD methods, but to assess whether the inclusion of ground-truth supervision acts as an additional leakage channel that increases the effectiveness of model inversion attacks against the distilled student model.

In this setup, we fix both the teacher and student to the **baseline** MLP and train the student using either *Traditional KD* or *Teacher-Only KD*. For each KD method, we sweep the temperature $T \in \{0.1, 0.5, 1.0\}$, and for Traditional KD we additionally vary $\alpha \in \{0.1, 0.5, 1.0\}$. Each (T, α) configuration yields a separate student, which we then attack using both MIA methods to reconstruct the teacher’s training data. We report attack accuracy, KNN distance, and attack utility averaged across all configurations and both MIA methods.

Table 3 indicates a higher privacy risk under Traditional KD. Specifically, Traditional KD yields an attack privacy accuracy of $59.19 \pm 5.53\%$, compared to $38.61 \pm 3.87\%$ for Teacher-Only KD. In contrast, KNN distance and attack utility are nearly identical for both KD methods. Overall, ground-truth supervision during distillation (Traditional KD) makes the student more vulnerable to MIA, suggesting that KD for IoT IDS applications should explicitly account for privacy to prevent exposure of the teacher’s sensitive data.

5.3 RQ3: Influence of teacher-student architecture on MIA effectiveness

Table 4. Average MIA performance of different methods across various teacher–student architectures.

Metric	Traditional MIA (mean $\pm$ std)	Improved MIA (mean $\pm$ std)	Comparison (Traditional MIA - Improved MIA)
Attack Accuracy (%)	67.80 $\pm$ 28.85	54.01 $\pm$ 30.74	+13.79
KNN Distance	5.9140 $\pm$ 0.40	5.9091 $\pm$ 0.41	+0.0049
Attack Utility (%) - mlp1hidden	31.61 $\pm$ 3.87	29.12 $\pm$ 4.19	+2.49
Attack Utility (%) - mlp4hidden	35.37 $\pm$ 7.26	31.01 $\pm$ 4.51	+4.36
Attack Utility (%) - random forest	24.29 $\pm$ 6.93	22.98 $\pm$ 7.59	+1.31
Attack Utility (%) - xgboost	29.43 $\pm$ 6.97	26.90 $\pm$ 7.69	+2.53

Table 5. Average MIA performance across various teacher–student architectures

Metric	Student (1024×1024)×1	Student (2048×2048)×1	Student (1024×1024)×2	Student (512×512)×2	Best Model
Attack Accuracy (%)	59.28 ± 38.32	65.17 ± 27.53	60.90 ± 27.36	58.27 ± 31.91	(2048 × 2048) × 1
KNN Distance	5.90 ± 0.43	5.91 ± 0.43	5.92 ± 0.41	5.92 ± 0.41	(1024 × 1024) × 1
Attack Utility (%) - mlp1hidden	30.52 ± 4.02	31.01 ± 3.92	30.13 ± 5.21	29.78 ± 4.16	(2048 × 2048) × 1
Attack Utility (%) - mlp4hidden	33.22 ± 6.60	34.84 ± 8.79	31.72 ± 5.83	32.97 ± 4.23	(2048 × 2048) × 1
Attack Utility (%) - random forest	24.01 ± 8.38	24.10 ± 7.72	22.98 ± 6.92	23.44 ± 7.05	(2048 × 2048) × 1
Attack Utility (%) - xgboost	28.20 ± 8.07	28.18 ± 7.83	27.78 ± 7.64	28.49 ± 7.28	(512 × 512) × 2

In practical deployments, the student often differs from the teacher (e.g., it is smaller or shallower) to satisfy resource constraints. We therefore investigate how teacher–student architectural discrepancies affect MIA effectiveness. In this setup, we fix the teacher to the **baseline** MLP and distill four student variants using *Traditional KD* with a fixed configuration ($T = 0.5$, $\alpha = 0.5$): (i) **baseline** $(1024 \times 1024) \times 2$, (ii) **shallow** $(1024 \times 1024) \times 1$, (iii) **wider shallow** $(2048 \times 2048) \times 1$, and (iv) **narrow** $(512 \times 512) \times 2$. For each trained student, we run both inversion methods (Traditional MIA and Improved MIA) and evaluate attack effectiveness using *Attack Accuracy*, *KNN Distance*, and *Attack Utility*. Table 4 aggregates results across student architectures for each attack method, while Table 5 reports results per student architecture, aggregated over attack methods.

We make two observations from Tables 4 and 5. First, Traditional MIA is more effective than Improved MIA: its attack accuracy reaches 67.80 ± 28.85 , compared to 54.01 ± 30.74 for Improved MIA. Traditional MIA also achieves higher attack utility for all evaluated downstream models. Notably, both attacks yield essentially the same **KNN Distance**. Second, **varying the student architecture within this MLP family has minimal impact on MIA effectiveness**. Across student architectures, attack accuracy and attack utility vary only modestly. Nonetheless, we observe that the *wider shallow* student $(2048 \times 2048) \times 1$ is the most vulnerable overall across all metrics.

6 Discussion

Despite the insights provided by our empirical study, this work is intentionally scoped in several respects. First, our evaluation focuses on MLP-based architectures, which are widely used in IoT intrusion detection. While this choice enables a controlled and interpretable analysis of the interaction between knowledge distillation and model inversion, future work should consider more expressive architectures (e.g., 1D-CNNs or Transformer-based models) to assess whether similar privacy-leakage patterns persist.

Second, the assumed attacker capabilities warrant further exploration. In this study, we assume white-box access to the distilled student model, reflecting a realistic edge-deployment scenario in which the student is accessible while the proprietary teacher remains protected. Extending this analysis to query-limited or strictly black-box inversion settings would provide a more comprehensive understanding of KD-related privacy risks in commercial environments.

Finally, model inversion on tabular network traffic raises challenges related to semantic validity. Even when numerical metrics are high, reconstructed samples may violate logical or protocol-specific constraints inherent to network features, reducing their practical exploitability. Incorporating domain-aware constraints and developing validity-aware evaluation metrics are important directions for future work, and will be crucial for designing robust, privacy-preserving distillation frameworks for cybersecurity applications.

7 Conclusion

This paper provides a systematic empirical analysis of privacy leakage induced by knowledge distillation in *tabular* intrusion detection systems under model inversion attacks. Our main contributions are threefold. First, we establish a strong tabular-domain baseline by benchmarking two representative inversion families (Traditional and Improved/logit-based) against multiple MLP teachers, showing that Traditional MIAs are consistently more effective in this setting. Second, we quantify how the distillation objective itself affects leakage by comparing Traditional KD and Teacher-Only KD on four widely used IDS datasets, and demonstrate that removing ground-truth supervision during distillation substantially reduces inversion success while largely preserving downstream utility of reconstructed data. Third, we investigate the influence of teacher–student architectural mismatch by distilling multiple student variants and show that, within the studied MLP family, architectural changes have only a modest effect on inversion effectiveness relative to the choice of KD and attack method. Overall, these results highlight an underappreciated privacy risk when releasing distilled student models for IoT NIDS, and provide actionable guidance: privacy-preserving distillation choices (e.g., Teacher-Only KD) are more impactful than minor architectural adjustments for mitigating inversion leakage in tabular IDS deployments.

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